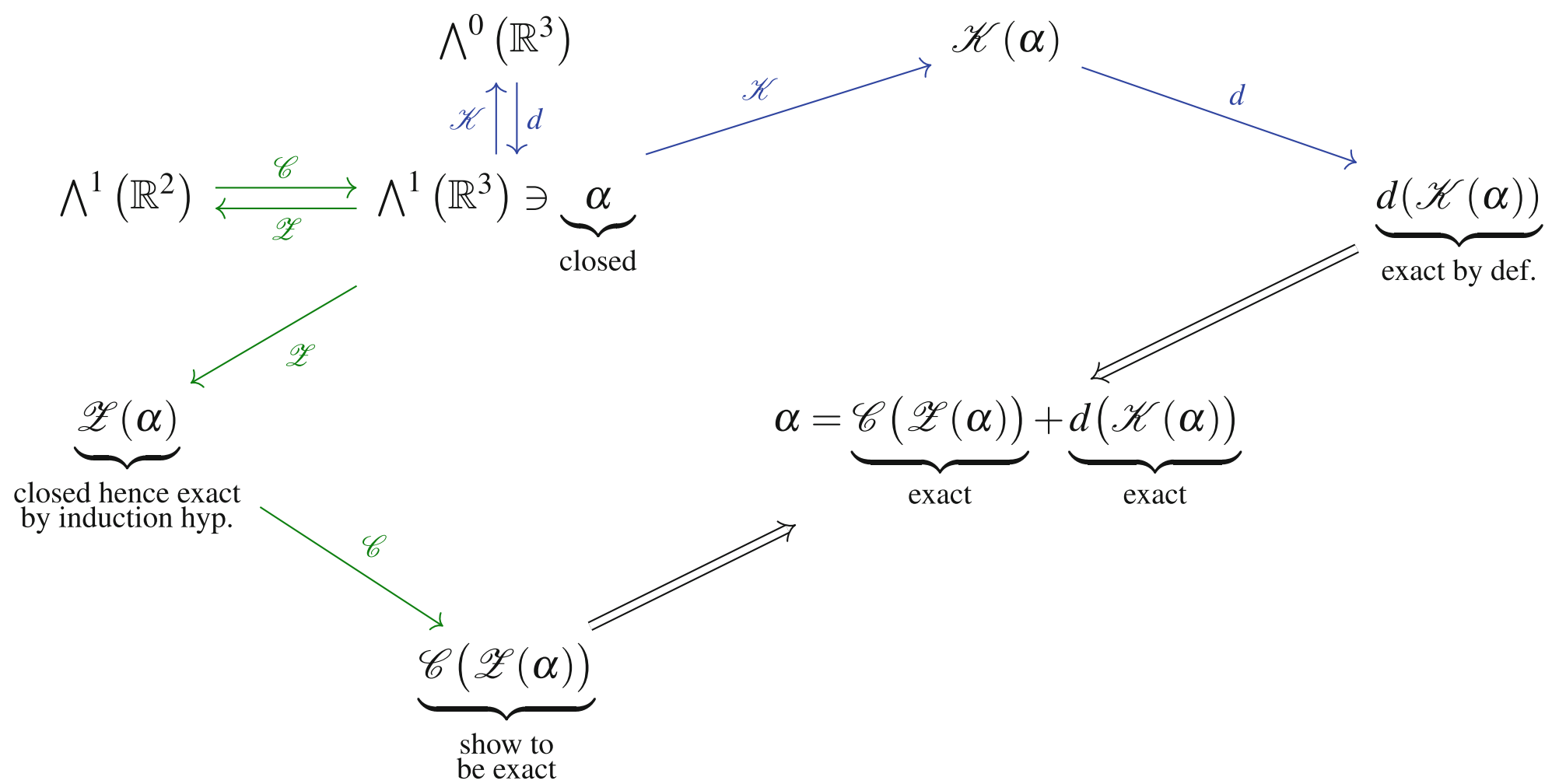

$\bigwedge^0(\mathbb{R}^3)$
$\mathscr{K}$
d
$\mathscr{K}(\alpha)$
$\mathscr{K}$
d
$\bigwedge^1(\mathbb{R}^2)$
$\mathscr{C}$
$\mathscr{Z}$
$\bigwedge^1(\mathbb{R}^3) \ni \underbrace{\alpha}$
closed
$\underbrace{d(\mathscr{K}(\alpha))}$
exact by def.
$\mathscr{Z}$
$\mathscr{Z}(\alpha)$
closed hence exact
by induction hyp.
$\alpha = \underbrace{\mathscr{C}(\mathscr{Z}(\alpha))}_{\text{exact}} + \underbrace{d(\mathscr{K}(\alpha))}_{\text{exact}}$
$\mathscr{C}$
$\underbrace{\mathscr{C}(\mathscr{Z}(\alpha))}$
show to
be exact